Enterprise Python Data API - Production Architecture

API/Control Plane
Processing Plane
Streaming/Queue
Storage Plane
Security/Identity
Observability

API Gateway (Traefik/NGINX)

FastAPI Application API (Control Plane)

IAM/RBAC | Dataset Registry | Job Management | Audit

Identity & Access

Keycloak + OIDC

Data Processing Plane

Python Workers (Celery)

Polars + PyArrow

Pandera (Quality)

Batch Processing

Real-time Processing

Redpanda/Kafka

Event Consumers

Job Queue

Celery + Redis

Storage Plane

PostgreSQL (Metadata/Jobs/RBAC/Audit)

S3-Compatible Object Storage (MinIO/S3)

Parquet + PyArrow (Raw/Curated)

Data Formats & Metadata

Observability

OpenTelemetry

Prometheus + Grafana

Tracing, Metrics, Logs

On-Premises - Private Data Zone

Private PostgreSQL (MinIO)

Sensitive Data Processing

Private Connectors

Site-to-Site VPN / Private Link (TLS + Firewall)

Cloud - Application & Elastic Processing Zone

API Gateway (FastAPI)

Elastic Workers (Scalable)

Event Broker